How to use: helper scripts

summarize_folder_structure.py · check_overdetection.py · restore_original_spots.py · check_restore.py. pipeline version 4 (October 2026).

All four are read-only except restore_original_spots.py, which only moves files (never deletes).

summarize_folder_structure.py

Reads folder names, file names and TIFF/CSV headers (never pixel data) and writes folder_structure_report.txt next to the script: folder tree, file-name patterns (digits → #), image size, pixel size, z-slices, software. Use REDACT_WORDS (e.g. {"partner": "PARTNER"}) before sharing the report.

check_overdetection.py

After Fish_spots_batch_v4.py. For every image: counted spots, median spot brightness, near-duplicate fraction, weak-spot fraction, each compared with the other images of the same pair. Output: _density_results/overdetection_check/overdetection_per_image.csv (flagged first, with likely cause) and overdetection_overview.png.

Flag	Threshold (setting)	Fix
dim puncta counted	brightness < 0.6× pair (DIM_RATIO_FLAG)	redetect with a pair-consistent threshold, or manual intensity cutoff
duplicate detections	> 10% spots closer than 250 nm	merge step (ask for it)
many weak spots	> 50% with SNR < 8	stricter SNR for that batch
many more spots	> 2× pair (COUNT_RATIO_FLAG)	inspect visually

Limitation: it compares images within a pair. If ALL images of a pair or batch are overdetected (as seen for the VisiView/EMCCD batch), nothing stands out - compare batches via counts and spot SNR, and check QC overlays.

restore_original_spots.py and check_restore.py

1. Run restore_original_spots.py with DRY_RUN = True: it writes restore_plan_DRYRUN_<date>.csv (one row per image: which original is restored, which batch file archived) - check it.
2. Set DRY_RUN = False and run again: originals come back from _old_spots; batch files, caches and v4 outputs go to _archive_batch_detection folders. Nothing is deleted or overwritten.
3. Run check_restore.py: restore_check.txt reports how many images have exactly one original spot file.